Serum concentrations of vitamin B12 and folate in British male omnivores, vegetarians, and vegans: results from a cross-sectional analysis of the EPIC-Oxford cohort study

. . . . .

Background/Objectives Vegans and to a lesser extent vegetarians have low average circulating concentrations of vitamin B12; however, the relation between factors such as age or time on these diets and vitamin B12 concentrations is not clear. The objectives were to investigate differences in serum vitamin B12 and folate concentrations between omnivores, vegetarians and vegans and to ascertain whether vitamin B12 concentrations differed by age and time on the diet. Subjects/Methods A cross-sectional analysis involving 689 men (226 omnivores, 231 vegetarians and 232 vegans) from the European Prospective Investigation into Cancer and Nutrition Oxford cohort. Results Mean serum vitamin B12 was highest among omnivores (281, 95% CI: 270-292 pmol/l), intermediate in vegetarians (182, 95% CI: 175-189 pmol/l), and lowest in vegans (122, 95% CI: 117-127 pmol/l). Fifty-two percent of vegans, 7% of vegetarians and one omnivore were classified as vitamin B12 deficient (defined as serum vitamin B12 < 118 pmol/l). There was no significant association between age or duration of adherence to a vegetarian or a vegan diet and serum vitamin B12. In contrast, folate concentrations were highest among vegans, intermediate in vegetarians, and lowest in omnivores, but only two men (both omnivores) were categorised as folate deficient (defined as serum folate < 6.3 nmol/l). Conclusion Vegans have lower vitamin B12 concentrations, but higher folate concentrations, than vegetarians and omnivores. Half of the vegans were categorised as vitamin B12 deficient and would be expected to have a higher risk of developing clinical symptoms related to vitamin B12 deficiency.